

HARAMAYA UNIVERSITY
COLLEGE OF AGRICULTURE AND ENVIRONMENTAL SCIENCES
SCHOOL OF AGRICULTURAL ECONOMICS AND AGRIBUSINESS
AGRICULTURAL ECONOMICS PROGRAM

Mock Examination

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Haramaya University

PART I: MULTIPLE CHOICE QUESTIONS (110)

INSTRUCTION: CHOOSE THE BEST CORRECT ANSWER FOR THE FOLLOWING QUESTIONS AND WRITE THE ANSWER ONLY ON PROVIDED SPACE

1. One of the following is not the major characteristics of a monopolistically competitive market
 - A. Product differentiation
 - B. Price competition
 - C. Free entry and exit of firms
 - D. Selling cost
2. The law of diminishing returns states
 - A. The total product may increase at first but will decrease after certain point
 - B. The total product may decrease at first but will increase after certain point
 - C. The total product always remain constant throughout all production stage
 - D. All
3. Which of the following statements is FALSE?
 - A. The types of goods and services offered by businesses are not influenced by what consumers are willing to pay for them.
 - B. Microeconomic principles tell us that businesses will set productivity at the highest level of economic efficiency.
 - C. When the price of a good increases, consumers have less utility available and may substitute another good or service.
 - D. Microeconomic principles tell us that businesses need to find a price point that maximizes efficiency but does not drive consumers to alternatives.
4. Mark has \$15 that he wants to spend on something that entertains him. He was planning to buy a book with this money, but then he decided to watch a new movie with this money. The fact that Mark cannot read the book because he opted for the film is an example of _____.
 - A. Opportunity cost
 - B. Elasticity
 - C. Equilibrium
 - D. Profit maximization

5. On a normal demand curve, if we know an individual will demand five units of a good when the good is priced at \$5, what statement is MOST LIKELY true if the price of the good changes to \$4?

- A. The individual will purchase fewer units of the good.
- B. A price change will cause a shift in demand and more of the good to be bought.
- C. The individual will purchase more units of the good.
- D. A movement along the demand curve will result in fewer units being purchased.

6. Which of the following allows us to cope with scarcity of resources?

- A. Increased efficiency, or getting more outputs from the same inputs
- B. Decreased utility, or reducing how useful the resource is to prevent it from being consumed
- C. Decreased demand, or reducing the number of people who want the scarce resource
- D. Increased supply, or creating more of the resource to reduce its scarcity

7. Suppose the first four units of a variable input generate corresponding total outputs of 200, 350, 450, 500. The marginal product of the third unit of input is:

- A. 50
- B. 100
- C. 150
- D. 200

Handwritten calculations for Question 7:

$$\begin{array}{r} 200, 350, 450, 500 \\ \hline 100, 100, 50, 25 \end{array}$$

$$\frac{2000}{1} = 2000$$

8. The production function for a product is given by $Q = 100KL$. If the price of capital is \$120 per day and the price of labor \$30 per day, what is the minimum cost of producing 1000 units of output?

- A. $TC = \$379.20$.
- B. $TC = \$381.60$.
- C. $TC = \$479.20$.
- D. NONE

Handwritten note: $Cost = Output \times Price$

9. Which one of the following is true?

- A. Increasing marginal utility, total utility increase at decreasing rate
- B. Decreasing marginal utility, total utility increase at increase rate
- C. Increasing marginal utility, total utility increase at increase rate
- D. All

10. The cross price elastic is negative:

- A. The products are independent each other
- B. The products are substitution each other

- C. The products are complementary each other
D. None
11. Which one of the following is **not** feature of monopolistic competitive market?
- A. There are many sellers and many buyers in a given market.
B. Producers produce dissimilar and differentiated products.
C. The goal of the firm is profit maximization only in the short run.
D. B and C are the answer.
12. Which of the following statement is **not** true?
- A. Monopolistic competitive is perfect competition because a large number of firms exist and entering the market is free.
B. Monopolistic competitive is monopolistic because each firm has some control over market conditions.
C. Monopolistic competitive is not a perfect competition because the products are not homogenous and the price is not uniform.
D. Monopolistic competitive is not a monopoly because the producer is not a single individual.
E. None of the above
13. In an oligopolistic market, there are:
- A. Large number of sellers and few buyers
B. Few sellers and few buyers
C. Few sellers and large number of buyers
D. Only one seller
14. Organization for Oil Producing and Exporting Countries (OPEC) is the best example of:
- A. Price Leadership
B. Cartel
C. Kinked demand model
D. Cournot's duopoly model
15. One of the following is correct when an employer has a monopoly power in the product market while it operates in a competitive labor market.
- A. Each unit of labor is paid a wage rate equal to the MRP_L .
B. The labor supply function of the firm is perfectly elastic.

C. The VMP_L is greater than the MRP_L .

D. A and C

E) All of the above

16. Which of the following is a wrong statement?

A. Both in perfectly competitive and imperfectly competitive markets, the demand for a single variable factor is the MRP_L .

B. Both VMP_L and MPP_L are decreasing functions of the level of labor employment.

C. The market supply curve of labor has a backward bending segment beyond a certain level of wage rate.

D. The extra cost of hiring one more unit of labor by a monopsonist of a single variable factor is higher than the wages of the additional unit of labor.

E. None of the above

17. Efficient allocation of a good with costs that are external to the production process occurs at:

A. Marginal private cost is equal to marginal private benefit.

B. Marginal external cost is equal to marginal social cost.

C. Marginal social cost is equal to marginal private benefit.

D. There is no efficient allocation of goods with external effect.

18. Public goods are provided by government because:

A. Private firms do not take into account the impact of external costs.

B. Governments are more efficient than private firms at producing public goods.

C. People value national defense very highly.

D. Private firms will make an economic profit.

E. Free-rider problems result in underproduction by private markets

19. Situation in which players make strategic decisions that take into account each other's actions and responses is known as:

A. Game theory B. Game C. Mixed strategies D. Payoff

20. If it is impossible to make anyone better-off without making someone worse-off, the situation is known as:

A. Pareto optimal B. General equilibrium C. Walrasian System D. None

21. All of the followings are there according to the classical economists' assumptions, Except

- A. Individuals are rational and markets are efficient
 - B. It has the invisible hand ruling principle ✓
 - C. Laissez-fair economy approach
 - D. Non-interventionist economy approach ✓
 - E. Government intervention is inevitable *possible*
 - F. Coined by economist such as Jean Say, David Ricardo and Adam Smith ✓
22. One of the following is false about National income accounting
- A. It is the process of record keeping the overall economic activities of a given country
 - B. It is the most comprehensive measure of how well an economy is performing
 - C. It is finding out the net output of a country through additions and deductions
 - D. It is measuring the value of economic activity such as GDP and GNP
 - E. None of the above
23. Which one of the following do not reflect Real Gross Domestic Product (RGDP)?
- A. It is GDP in a selected base year price ✓
 - B. It is GDP during production period price
 - C. It is GDP in constant price
 - D. It is GDP adjusted for inflation
24. Which of one of the following is false about Aggregate demand
- A. It is the total amount of goods and services demanded in the economy at a given overall price level in a given time period.
 - B. It shows the level of real GDP at different possible price levels during a given time period.
 - C. It has a downward slope curve which depicted the relationship between the general price level and the amount of real GDP demanded
 - D. All are correct
25. A situation in which the problem of inflation arises along with low aggregate output or during economic sluggishness or recession is known as *low out*
- A. Cost - Push Inflation
 - B. Demand - Pull Inflation
 - C. Stagflation
 - D. Pure Inflation

E. None

26. Which one of the following is not the concern of Macroeconomics?
- A. It focuses on the economic behavior and policies that affect consumption and investment ✓
 - B. It concerned with the behavior of the economy as a whole- with booms and recessions ✓
 - C. It deals with both long-run economic growth and the short run fluctuations which are constituents of the business cycle.
 - D. It concerned with the determinants of changes in wages and prices, monetary and fiscal policies including trade balance and national debt
 - E. None of the above
27. According to the New Keynesian economists, prices and wages are not flexible as opposed to the classical economists' due to the following major reasons, Except
- A. Existence of strong trade unions and long-term contract
 - B. Existence of perfect information for market agents
 - C. Existence of minimum wage rate legislation
 - D. Existence of monopoly power of labour unions
 - E. None of the above
28. One of the type of unemployment arises from insufficient aggregate demand in the economy to generate enough jobs for those seeking them and known as demand deficient unemployment
- A. Structural unemployment
 - B. Seasonal unemployment
 - C. Cyclical unemployment
 - D. Frictional unemployment
29. One of the following monetary policy tools is applicable by the central bank to increases the money supply in the economy, Except?
- A. A decrease in reserve requirements ✓
 - B. The central bank buys bonds in the nation's bond markets
 - C. The central bank imposes a lower discount rate for banks
 - D. The central bank ^{central bank} sells government bonds to the public in the nation's bond markets
30. Which one of the following is correct about "tight money" monetary policy?
- A. It is an expansionary monetary policy

- B. It is a contractionary monetary policy
- C. It involves lowering interest rates and increasing the money supply
- D. It involves raising interest rates and decreases the money supply
- E. A and C
- F. B and D

31. In Keynesian model, an economy's total income in the short run, determined largely by:

- A. The desire to spend by households, firms and the government
- B. The stock of additional capital or the net investment in the economy
- C. Government influence exerted in production and consumption
- D. The competitiveness of the market in the economy

32. Which one of the following theory of investment describe that investment is based on benefit and cost of the investment activity to a firm or firms and the level of investment has to keep on increasing as long as its benefit is greater than the cost of the investment activity?

- A. Accelerator Theory of Investment
- B. Keynesian marginal efficiency of capital (MEC)
- C. Neo-classical theory of investment
- D. Internal fund theory of investment

33. One of the following monetary policy instrument deal with the purchase and sale of government securities by the central bank to maintain the proper level of money supply in an economy

- A. Reserve requirements
- B. Open market operation
- C. Changes in the discount rate
- D. Tax and government expenditure

34. Which one of the following can be a cause for appreciation of domestic currency?

- A. An increase in demand^{des} for a foreign currency relative to its supply
- B. A decrease in demand^{des} for a foreign currency relative to its supply
- C. An increase in supply of a foreign currency relative to its demand
- D. A and B
- E. B and C

35. Which one of the following theories of demand for money explains that money is demanded to facilitate transaction, for speculative purpose and as security against unforeseen needs for cash reserves which is called precautionary demand for money?
- Classical theory of money demand
 - Keynesian theories of money demand
 - Portfolio theory of money demand
 - None of the above
36. Assume that an Ethiopian citizen firms done an investment in foreign securities or stocks or bonds or shares in USA, this type of long-term capital account is considered as
- Private direct investment and considered as ~~debit~~ ^{credit} in Ethiopian Balance of Payment
 - Private portfolio investment and considered as credit in Ethiopian Balance of Payment
 - Private direct investment and considered as credit in Ethiopian Balance of Payment
 - Private portfolio investment and considered as debit in Ethiopian Balance of Payment
37. Which one of the following reflects a possible shift of the IS curve to the leftward?
- An increase in government expenditure and a reduction in tax
 - A decrease in government expenditure and an increase in tax
 - When AD is greater than the full employment level of income and government apply contractionary fiscal policy
 - When there is high unemployment in the economy and government apply expansionary fiscal policy
 - A and D
 - B and C
38. Suppose the aggregate demand function of a country's closed economy is given by consumption function, the investment and government expenditure as follows: $C=100+0.75Y$; $I=150$ and $G=500$. Then the equilibrium level of income and consumption respectively becomes
- | | |
|------------------|------------------|
| A. 5000 and 1500 | B. 3000 and 2350 |
| C. 4500 and 850 | D. 2350 and 1250 |
39. Surplus in overall balance of payments is a good sign in the overall balance of payments of a country in one of the following conditions
- If the overall surplus in the BOP was caused by capital account surplus

- B. If the overall surplus in the BOP was caused by current account surplus
- C. If the overall surplus in the BOP was earned from Short term capital movements
- D. If the overall surplus in the BOP was earned from Smuggling
- E. None of the above

40. In which one of the following exchange rate policy does the exchange rate is dependent upon the supply and demand of money in the market?

- A. Flexible exchange rate policy
- B. Fixed exchange rate policy
- C. Floating exchange rate policy
- D. Domestic exchange rate policy
- E. A and C

41. Which of the following is/are type of Non-use value

- A. Option value
- B. Existence value
- C. Bequest value
- D. B & C
- E. A & B

42. Which one of the following is true about Instruments for achieving pollution abatement targets

- A. Marketable permit systems are based on the principle that any increase in emissions must not be offset by an equivalent decrease elsewhere satisfying pareto improvement.
- B. A cap-and-trade marketable emission permits scheme for a uniformly mixing pollutant in which the cap is the total amount of permits issued (measured in units of pollution) that should be equal to that target level of emissions.
- C. Marketable permit schemes differ from tax or subsidy schemes by working in terms of quantities rather than prices
- D. None of the above

43. Which of the following is/are type of Discrete Choice Survey

- A. Payment card Format
- B. Referendum Format
- C. Bid Value

D. All

44. Sustainable development means maximizing

- A. Economic growth
- B. Income inequality
- C. The productivity of produced assets
- D. None of the above

45. Which of the following is/are the type for No Proxies

- A. Contingency Valuation Method
- B. Hedonic Pricing
- C. Reveled Preference Method
- D. Travel Cost Method
- E. B&D
- F. A&C

46. For markets to produce efficient allocations, it is necessary that:

- A. Markets exist for all goods and services produced and consumed
- B. Private property rights are fully assigned in all resources and commodities.
- C. All markets are perfectly competitive
- D. All

47. Which of the following is/are approach Hedonic Pricing Method

- A. Wage differential approach
- B. Payment vehicle approach
- C. Interviewer approach
- D. None of the above

48. One of the following is incorrect statement about externalities

- A. In the case of positive externality, an unregulated market system would over produce (desired) quantity of a given good
- B. In the case of negative externality, an unregulated market system would under produce the [undesired] quantity of a give good
- C. Externalities are intended outputs of an economic activities wher compensations are paid
- D. None

- E. All
49. Which of the following is/are the type of Indirect Proxies.
- A. Contingency Valuation Method
 - B. Hedonic Pricing
 - C. Reveled Preference Method
 - D. Travel Cost Method
 - E. B&D
 - F. A&C
50. Determinants of environmental degradation includes all of that except:-
- A. Market failure.
 - B. Defective economic policies.
 - C. Defined property rights.
 - D. Tragedy of commons.
51. Which one of the following is not a goal of econometrics?
- A. Developing economic theory
 - B. Developing economics models
 - C. Testing economic models
 - D. Utilizing economic models
 - E. None of the above
52. Econometric model is different from mathematical model since the former
- A. Involves stochastic error terms
 - B. Involves linear models
 - C. Has no deterministic nature
 - D. Has no its own testing technique
 - E. All of the above
53. Which of the following Z-value shows that the explanatory variable included in to the model has a significant effect on the dependent variable with 5% level of significance?
- A. 0.501
 - B. 0.100
 - C. 0.018
 - D. 0.051
54. Assume that the simple correlation between two variables was found to be 0.835, then which one of the following is true about the correlation of these two variables?
- A. The two variables have a strong positive correlation
 - B. They do have a weak positive correlation
 - C. The two variables have a perfect positive correlation
 - D. They do not have any correlation

55. Let's assume that two individuals were allowed to give a rank for two brands of soap. The rank correlation between the two brands of soap was found to be 0.75 then which one of the following shows the correct interpretation for the result obtained?

- A. The two persons have the marked similarity of 0.75 in choosing the two brands of soap
- B. As the number of soap A purchased increase by one unit then the number of soap B purchased will decrease by 0.75
- C. The two persons have the marked difference of 0.75 in choosing the two brands of soap
- D. As the number of soap A purchased increase by one unit then the number of soap B purchased will increase by 0.75
- E. None of the above

56. Imagine that you obtained the following multicollinearity test result using VIF test.

Variable	VIF
Initial bid	2.51
Years lived in the area	2.00
Age of the HH	1.93
Farm experience	1.90
Distance from the market	1.43
Total non-farm income	1.32
Size of own land	1.25
Farm income	1.21
Family size in AE	1.19
Dependency ratio	1.17
Frequency of extension contact	1.15
Education level of the HH	1.13
Mean VIF	1.52

Then which one of the following is true about the test?

- A. There is no any multicollinearity in the data set
- B. There is serious multicollinearity in the data set
- C. There is multicollinearity but not serious
- D. The test proves that multicollinearity is not an inherent phenomena

***Answer the following questions from 57 to 60 based on the given summary results.**

$$\sum Y = 639$$

$$\sum X_1 = 3679$$

$$\sum X_2 = 1684$$

$$n = 12$$

$$\bar{Y} = \frac{\sum Y}{n} = \frac{639}{12} = 53.25$$

$$\bar{X}_1 = \frac{\sum X_1}{n} = \frac{3679}{12} = 306.5833$$

$$\bar{X}_2 = \frac{\sum X_2}{n} = \frac{1684}{12} = 140.3333$$

The following summary results in deviation forms are also given:

$$\sum x_1^2 = 49206.92$$

$$\sum x_2^2 = 2500.667$$

$$\sum x_1 y = 1043.25$$

$$\sum x_2 y = -509$$

$$\sum x_1 x_2 = 960.6667$$

$$\sum y^2 = 1536.25$$

57. By using the above given, which one of the following shows the correct value of the coefficients?

A. $\hat{\beta}_1 = 0.0254; \hat{\beta}_2 = -0.213$

B. $\hat{\beta}_1 = 75; \hat{\beta}_2 = -0.213$

C. $\hat{\beta}_1 = 0.0254; \hat{\beta}_2 = 75$

D. None of the above

58. The adjusted R^2 value and its correct interpretation was:

A. $R^2 = 0.08789$ (as X 's increases by one unit the dependent variable Y increases by 8.789%)

B. $R^2 = -0.1148$ (as X 's increases by one unit the dependent variable Y decreases by 11.48%)

- C. $R^2 = 0.08789$ (the explanatory variables included in to the model was explained or determined the dependent variable by 8.789%)
- D. $R^2 = -0.1148$ (the explanatory variables included in to the model was explained or determined the dependent variable by 11.48%)
- E. None of the above

59. Additional Given: $\text{Var}(\hat{\beta}_1) = 0.0032$; $\text{Var}(\hat{\beta}_2) = 0.063$ and $t\text{-tabulated} = 2.262$. Then choose the letter that holds the correct test result using student t-test for the coefficient $\hat{\beta}_1$ and the decision.

- A. The t-calculated value is 0.44925 and the decision is to reject the null hypothesis
- B. The t-calculated value is 2.3125 and the decision is to reject the alternative hypothesis
- C. The t-calculated value is 0.44925 and the decision is to accept the null hypothesis
- D. The t-calculated value is 2.3125 and the decision is to accept the alternative hypothesis
- E. None of the above

60. Additional Given: The F-tabulated value (critical value) is 3.98. Then by using all the required givens from the above, the overall significance of the model (F-calculated value) and the associated decision was found to be:

- A. 0.4336 and reject the null hypothesis
- B. 4.336 and accept the alternative hypothesis
- C. 4.336 and reject the alternative hypothesis
- D. 0.4336 and accept the null hypothesis
- E. None of the above

61. Which one of the following is **not correct** definition of research in social sciences?

- A. Research is structured enquiry that utilizes acceptable scientific methodology to create new knowledge for phenomenon.
- B. Research is considered as a movement from the known to the unknown for making deductions and reaching conclusions.
- C. Research refers to the systematic method consisting of enunciating the problem and analyzing the facts for making certain conclusions in the form of solutions.
- D. None

62. Which one of the following is the **correct** statement about types of researches?

- A. Fundamental research aims at finding a solution for an immediate problem confronting a society in certain phenomenon.

- B. Descriptive research is ex post facto research in which the researcher has control over the variables
 - C. Qualitative research is aim to discover the underlying motives of human behavior and to find out how people feel about a particular subject.
 - D. Conceptual research is data-based research coming up with conclusions which are capable of being verified by observation or experiment.
63. Suppose epidemic disease covid-19 has been affecting several hundred peoples in Hawassa city, Ethiopia. What type of research would be employed to find an immediate solution to solve the prominent problem in the society?
- A. Applied Research
 - B. Basic Research
 - C. Conceptual Research
 - D. Explanatory Research
64. Which one of the following is **correct** statement about application of sample size determination formula in the research study?
- A. Yemane formula applied when the population under study is finite and heterogenous
 - B. Kothari formula applied when the population under study is finite and heterogenous
 - C. Cochran formula applied when the population under study is finite and homogenous
 - D. None
65. Which one of the following could not be considered while making extensive literature survey?
- A. Academic journals
 - B. Conference proceedings
 - C. Government reports
 - D. Field survey
66. Which one of the following is **not** importance of developing working hypothesis in research study?
- A. It is a tentative assumption made in order to draw out and test its empirical consequences
 - B. It is to guide the researcher by limiting the area of research and to keep him on the right track
 - C. It sharpens his thinking and focuses attention on the more important facets of the problem
 - D. It indicates the type of data required and the type of methods of data analysis to be used
67. Which one of the following is the **correct** statement about probability sample design in research study?
- A. Systematic sampling is select some random starting point and then a systematic pattern is used in proceeding through the table
 - B. Stratified sampling is applied if the population from which a representative sample is to be drawn does constitute a homogeneous group
 - C. Cluster sampling involves grouping the population in to cluster then selecting individual elements for inclusion in the sample from each cluster
 - D. Simple random sampling can put the names of all the items on slips of paper and conduct a lottery to select representative sample from a finite population
68. Which one of the following is **correct** about primary data collection tools in conducting scientific research?

- A. Observation implies the collection of information by way of investigator's interviewing the respondents
- B. Personal interview follows a flexible procedure to seek answers for a set of pre-conceived questions
- C. Schedule involves appointing enumerators filling up the schedules on the basis of replies given by respondents
- D. Questionnaires are mailed to the respondents with a request to return after completing filling the required data.

69. Which one of the following is **not correct** statement about non-probability sample design in research study?

- A. Snowball sampling is applied to locate through referral networks where respondents are difficult to identify
- B. Purposive sampling involves deliberate selection of particular units of the universe for constituting a sample which represents the universe
- C. Judgement sampling is used for selecting items which he considers as representative of the population
- D. Quota sampling follow identifying groups in which elements are sampled from each group through random sampling

70. Determine the size of the sample for estimating the true weight of the cereal containers for the universe with $N = 5000$ on the basis of the following information:

- (1) the variance of weight = 4 ounces on the basis of past records.
- (2) estimate should be within 0.8 ounces of the true average weight with 99% probability
- (3) $z = 2.57$

How much the sample size will be? using kothari formula

- A. 82
- B. 53
- C. 41
- D. 32

71. Which assumption of linear programming problem shows the existence constant returns to scale or no economies of scale?

- A. Additivity assumption
- B. Finiteness assumption
- C. Proportionality assumption
- D. Divisibility assumption

72. Which one of the following is not the basic assumptions of linear programming?

- A. Certainty assumption
- B. Optimality assumption
- C. Single objective function
- D. Additivity assumption
- E. None of the above

73. Among the following one is not the components of linear programming problem:

- A. Controllable variables
- B. Uncontrollable variables
- C. Objective function
- D. Constraints or limitations
- E. None of the above

74. All of the following are limitations of operations research **except**:

- A. Operations research provides solution only when all elements related to a problem can be quantified.
- B. Quantifying and establishing relationships among many factors require voluminous calculations which can be handled by computers.
- C. Operations research being specialist's job requires a mathematician or statistician who might not be aware of the business problems.
- D. Operations research has practical utility limitation

75. Which one of the following is **not true** about the multiple optimal solutions?

- A. Each of the optima would naturally yield the same objective function value.
- B. The objective function should be parallel to some edge of constraint which is also boundary to the feasible region.
- C. The constraint that is parallel to objective function must be a binding constraint
- D. None of the above

A firm wants to maximize its profit by producing products, A and B. Letting X_1 as units of product A to be produced while X_2 as the units of product B, the following optimal simplex table of a given problem is obtained. Using this table try to answer question number 6 and 7.

CB	BV	40 X_1	80 X_2	0 S_1	0 S_2	0 S_3	Xb
40	X_1	1	0	1/2	0	-3/2	9
0	S_2	0	0	-1/2	1	3/2	6
80	X_2	0	1	0	0	1	10
	Z_j	40	80	20	0	20	1160
	Δ_j	0	0	20	0	20	

76. Which resources are full utilized?

- A. The first and third resources are full utilized.

- B. The first and second resources are full utilized.
 C. The second and third resources are full utilized.
 D. All resources are full utilized.

77. What is the value of objective function, if the firm adds 10 more units of the first resource?

- A. 1160 B. 1360 C. 1180 D. 23200

78. A company makes two kinds of leather belts. Belt A is a high-quality belt and belt B is of lower quality. The respective profits are Birr 4 and Birr 3 per belt. The production of each of type A requires twice as much time as a belt of type B, and if all belts are of type B, the company could make 1000 per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 per day are available. There are only 700 buckles a day available for belt B. **Based on the above problem**, let X_1 = number of belt A to be produced and X_2 = number of belt B to be produced, **then which linear programming formulation is the correct?**

A. $\text{Max } Z = 4X_1 + 3X_2$

Subject to

$$2X_1 + X_2 \leq 1000$$

$$X_1 + X_2 \leq 800$$

$$X_1 \leq 400$$

$$X_1 \leq 700$$

$$X_1 \geq 0, X_2 \geq 0$$

B. $\text{Max } Z = 4X_1 + 3X_2$

subject to

$$X_1 + X_2 \leq 800$$

$$X_1 \leq 400$$

$$X_2 \leq 700$$

$$X_1 \geq 0, X_2 \geq 0$$

C. $\text{Max } Z = 4X_1 + 3X_2$

Subject to

$$X_1 + 2X_2 \leq 1000$$

$$X_1 + X_2 \leq 800$$

$$X_1 \leq 400$$

$$X_1 \leq 700$$

$$X_1 \geq 0, X_2 \geq 0$$

D. $\text{Max } 3X_1 + 4X_2$

Subject to

$$X_1 + 2X_2 \leq 1000$$

$$X_1 + X_2 \leq 800$$

$$X_1 \leq 400$$

$$X_1 \leq 700$$

$$X_1 \geq 0, X_2 \geq 0$$

79. _____ is a type of non-linear programming problem used in the problem of portfolio selection with risky securities.

- A. Separable programming

- B. Linearly constrained Optimization
- C. Quadratic programming
- D. Convex Programming

80. Method of efficiency analysis which requires data on quantities produced and inputs used, but not on market prices is known as:

- A. Stochastic frontier approach
- B. Data envelopment analysis
- C. Distribution free approach
- D. A and C are the answer

81. One of the following is **not correct** about the theories of international Trade.

- A. The Heckscher-Ohlin trade theory asserts that each country should specialize in a commodity whose production requires intensive use of its scarce factor of production.
- B. The Heckscher-Ohlin trade theory asserts that each country should specialize in a commodity whose production requires intensive use of its abundant factor of production.
- C. According to absolute advantage principle, absolute cost difference between trading partners is the basis for international trade
- D. The comparative advantage principle asserts that relative cost difference between trading partners is the basis for international trade.

82. In a world of two factors of production, labor and capital, one of the following is **not correct** about Hicksian technological progress?

- A. Neutral technical progress increases the productivity of neither labor and capital.
- B. Labor-saving technical progress increases the productivity of capital proportionately more than the productivity of labor.
- C. Capital saving technical progress increases the productivity labor proportionately more than the productivity of capital.
- D. A country can produce a given output with fewer units of labor and capital as result of technical progress.

83. One of the following is **correct** about static effects of regional trade arrangements.

- A. Trade creation occurs when domestic production of one member is replaced by another member's lower-cost imports.

- B. Trade creation expands the volume of trade among the member countries.
- C. Trade diversion is a welfare loss resulting from the formation of a regional trade arrangement.
- D. The static welfare effect of regional trade arrangement is determined by the magnitude of trade creation and trade diversion.
- E. All are correct

84. Identify the **wrong** statement about export subsidy.

- A. It is designed to increase the volume of trade.
- B. Consumers in the exporting country suffer as international terms of trade moves against them.
- C. Consumers in the exporting country suffer as international terms of trade moves against them
- D. It includes especial tax exemptions and provision of capital at lower interest rates to help domestic producers of export commodity.

85. If the consumption effect of imported commodity is protrade and the production effect is neutral, what would be the overall effects of economic growth on trade?

- A. Protrade
- B. Antitrade
- C. Neutral
- D. Impossible to determine

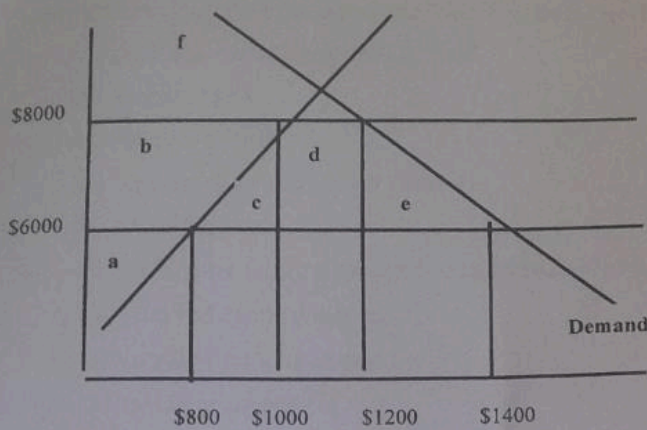
86. Identify the **wrong** statement about autarky equilibrium.

- A. A country can consume beyond its production possibility frontier. ✓
- B. A country's production is limited by its production possibility frontier. ✓
- C. A country's consumption is limited by its production possibility frontier.
- D. None

Suppose a small country imposed a specific tariff of \$2000 per TOYOTA Corolla car. Before a tariff the world price was \$6000 per TOYOTA Corolla car. Free trade production and consumption of TOYOTA Corolla were 800 and 1400, respectively. Now, due to the tariff levied, production is increased to 1000 while consumption declined to 1200 as shown in

Figure 1. Based on this answer questions 7-9.

Supply



87. The government of tariff imposing country collects "tax revenue" equivalent to which area?
- Area "a"
 - Area "e"
 - Area "d"
 - Area "f"
88. One of the following is **correct** about the effects tariff imposed by this small country.
- Producers welfare decreases by area "b".
 - Consumers welfare loss to producers is equivalent to area "b"
 - Producers welfare increases by area "f".
 - There is no welfare change on both producers and consumers
89. What is the total welfare effect of this tariff on the tax imposing country?
- \$200,000
 - \$200,000
 - \$400,000
 - \$400,000
90. The overall effect of regional trade arrangements is influenced by?
- The number of countries forming regional trade arrangement.
 - The competitiveness of countries forming regional trade arrangement.
 - The extent of retaliation by non-member countries.
 - All

91. Agriculture is important for one of the following in the developing countries
- A. As a sources of livelihood or food
 - B. Employment
 - C. Sources of GDP
 - D. Sources of industrial raw materials
 - E. All of the above
92. One of the following farming system is characterized consumption dependent
- A. Specialized farming system
 - B. Diversified farming system
 - C. Mixed farming system
 - D. Marginal farming system
 - E. All of the above
93. One of the following is the components of farming except:
- A. Crop production
 - B. Dairy farming
 - C. Poultry farming
 - D. Apiculture
 - E. None of the above
94. Most of the time the farming system of developing countries can be affected by;
- A. Science and technology
 - B. Natural resource and climate
 - C. Information and communication
 - D. Policy, institution and public
 - E. All of the above
95. Identify which term best describes all activities undertaken to produce an output that contributes to total production or income of the farm family.
- A. Off-farm activities
 - B. On-farm activities
 - C. Enterprise
 - D. All except A
 - E. None
96. Which of the following is the characteristic of farming system research approach?
- A. Farmer oriented

- B. Holistic
 - C. Problem solving
 - D. Up down approach
 - E. All except D
97. One of the following is true about the features of farming system research approach
- A. Identify the existing problems and opportunities of farming system
 - B. Carefully analyzes farm households through close and frequent contacts with farm family
 - C. Take in to account the marketing condition and public policy
 - D. Interested in the farm household heads' difficulty of farming
 - E. All but D
98. One of the following development model assumes expansion of cultivated and grazing land as a main source of growth
- A. Technology transfer model
 - B. Frontier model
 - C. Conservation model
 - D. Industrial impact model
 - E. None of the above
99. What is a livelihood intervention?
- A. It is an income generation program
 - B. It is the provision of food, water, health and shelter
 - C. Food for work programs
 - D. Improving production and marketing activities of households
 - E. All
100. _____ are pressures which are typically continuous and cumulative, predictable and distressing, such as seasonal shortages, rising populations or declining resources.
- A. Shocks
 - B. Protection
 - C. Assets
 - D. Stresses
 - E. None
101. It arises from the development and adoption of new techniques or methods of production.
- A. Institutional risk
 - B. Price or marketing risk

- C. Production risk
D. Techniques risk
E. Technological Risk
102. Which one of the following is true objective farm planning except;
A. There is underutilization and over utilization of the existing farm resources.
B. usually farm organizations fail to get maximum expense
C. for proper allocation of resources to obtain optimum production and net income
D. All
103. The difference between the gross receipt and total cost of production
A. Balance sheet
B. Net worth
C. Solvency
D. Liquidity 4.
104. It is the financial estimate of the annual loss of value of capital equipment due to wear and tear over its useful life.
A. Receipts
B. Expenses
C. Depreciation
D. Cost or market price
E. All
105. One of the following is not sources of land
A. Owning land
B. Leasing, Renting
C. Buying
D. All
E. None
106. One of the following is not valuation methods:
A. Net selling price
B. Cost less depreciation
C. Cost or market price (whichever is lower)
D. Income capitalization
E. None
107. Which one is different types of efficiency measures
A. Physical efficiency measures
B. Value efficiency measures
C. financial efficiency measures
D. None
108. Which one of the following is true about attitude towards risk?

- A. Risk averse person prefers to achieve the maximum output under risky condition
 - B. Risk averse person is a person who doesn't care whether risk occur or not
 - C. Risk taker person usually prefers to use maximax rule for making decision
 - D. All are correct
 - E. None
109. Which one of the following statement differentiates Risky situation from uncertainty?
- A. Uncertainty leaves the decision maker with no or imperfect information to set probability of occurrence of possible outcomes
 - B. The decision maker will not have enough information about the probability of occurrence of certain situation in case of risky situation
 - C. Possible outcomes of a certain situation is unknown in risky situation
 - D. All except A
 - E. None
110. One is different Characteristics of Land
- A. Land is permanent resource that does not depreciate
 - B. Land is mobile and can move to combined with other resources
 - C. The supply of land suitable for agricultural production is essentially fixed
 - D. None